Identity, Market Access, and Demand-led Diversification

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Paper Summary
This is an interactive guide addressing key concerns and highlighting contributions.

1 Summary

Core Research Question: How costly is identity-based market segmentation, and who gains from removing it? Which channel–demand-side consumer preferences or supply-side hiring barriers–matters more?

Main Finding: Firms hire workers from other caste groups not despite identity barriers but because doing so allows them to access consumers who would otherwise be unreachable–a mechanism we call *demand-led diversification.* A quantitative trade model separating demand-side taste bias from supply-side hiring frictions shows that supply-side frictions are the dominant source of misallocation, that the two channels are complementary, and that the burden falls disproportionately on the poorest group.

Key Quantitative Results:

- **21.5%** average ad valorem barrier from caste-based frictions–comparable to interstate trade costs in India
- **4.7%** aggregate income gain from removing supply-side hiring wedges alone
- **2.6%** aggregate income gain from removing demand-side taste bias alone
- **7.6%** combined gain from removing both (the two channels are complementary)
- **46%** reduction in cross-caste inequality when both frictions are removed

Question

"Is this just about India-specific caste issues?"
While we use India's caste system as a natural laboratory, our framework applies wherever social identity segments product markets–along ethnic, religious, racial, or sectarian lines. The mechanism requires: (1) consumers have identity preferences, (2) employee characteristics affect customer perceptions, and (3) firms can diversify their workforce. Evidence of similar patterns exists in the US (racial preferences on eBay, Airbnb), Sub-Saharan Africa (gender-based customer discrimination), and financial markets (social connections and lending). The theoretical framework generalizes to any form of social identity.

2 Research Design & Identification

2.1 Causal Identification Strategy

Challenge: Consumer demand and a firm's market access and hiring are endogenously determined.
Solution: Rainfall shocks as exogenous demand shifters that differentially affect caste groups.

How Rainfall Creates Identification:

1. **Asymmetric Income Effects:** LC households are poorer (demand is elastic to income shocks) <u>and</u> 59% work in agriculture ⇒ benefit more from good rainfall
2. **Demand Composition Shift:** LC household consumption increases 9.2% relative to HC households after a positive rainfall shock
3. **Firm Response:** LC-owned firm revenues rise by 4.7% relative to HC-owned firms; HC-owned firms in competitive markets respond by hiring more LC employees

Empirical Specification:

$$\ln y_{idt} = \rho_s \cdot \text{Rshock}_{dt} \times \mathbf{1}[\text{caste} = s] + \alpha_{dt} + \alpha_{pt} + \gamma_s + \varepsilon_{idt}$$

where ρ_{LC} identifies the differential revenue effect for LC-owned firms relative to HC-owned firms, with district×year and product×year fixed effects absorbing common shocks.

Question

"Aren't rainfall shocks just supply shocks to agriculture?"
We address this through multiple tests: (1) Non-agricultural wages do not respond to rainfall shocks (Table A.11), while agricultural wages do–confirming the demand channel operates through income, not labour supply shifts; (2) Effects are strongest in non-tradable goods and high-consumption-elasticity sectors; (3) The competition and contact-intensity interactions (Table 4) show that the hiring response varies with *product-market structure*, not district-level labour supply; (4) A foreign demand placebo–caste-neutral export shocks–does not generate differential LC hiring, ruling out the pure cost-based explanation.

2.2 Unique Dataset

MSME Survey 2006–07: 444,574 registered rural firms employing ∼3 million workers across 195 sectors and 4,590 products.

Data Advantage: The MSME survey is the only nationally representative dataset recording caste information for both employers *and* employees–unavailable in other major Indian firm datasets (ASI, Prowess). A key feature is that 99% of firms operate in overlapping sectors where all three caste groups compete.

Question

"Why focus on small/medium firms?"
Small/medium firms are where identity matters most–they operate in local markets with face-to-face transactions where seller identity is observable. Large formal-sector firms (where consumers cannot observe workforce caste) are less subject to demand-side identity bias. Moreover, MSMEs are the second-largest source of employment after agriculture ($\sim$30% of GDP). Our results are stronger among larger firms within the sample, contradicting simple selection stories.

2.3 Robustness Checks

Alternative Explanations Tested & Ruled Out:

- **Geographic Segregation:** Results hold in geographically unsegregated districts (using Census-based isolation index)
- **Product Quality:** Effects persist when restricting to products with low output and input price dispersion across castes
- **Supply-Side Labour:** Effects concentrated in competitive and contact-intensive sectors; foreign demand placebo shows no differential hiring
- **Financial Constraints:** Effects are stronger among larger (less constrained) firms
- **Price Discrimination:** Revenue effects are not driven by differential output or input prices

Question

"Could this just be wealth-based segregation disguised as caste preferences?"
We control for household wealth (land ownership, meals per day, education) interacted with rainfall. The caste effect remains. We also restrict to product markets where LC and HC firms use similar-quality inputs and produce at similar prices–results are quantitatively similar or stronger. The model's non-homothetic structure explicitly allows income differences to operate through market size, while the taste parameter β captures residual demand segmentation that income alone cannot explain.

3 Key Empirical Findings

Fact 1: Cross-Caste Hiring Is Prevalent but Asymmetric
38% of firms employ at least one worker from another caste. Own-caste workers account for 50–64% of employment–exceeding a random-hiring benchmark by 13–29 percentage points. HC workers account for 30% of employment in LC-owned firms, while LC workers account for only 20.5% in HC-owned firms.

Fact 2: Larger Firms Have Diverse Workforces
Own-caste hiring share falls monotonically from ∼85% (smallest firms) to ∼58% (largest firms). The size–diversity gradient is steeper in contact-intensive sectors (where customer interaction is salient) and among LC-owned firms (whose small home market pushes them into out-group markets).

Fact 3: Employee Composition Tracks Consumer Demand
Workforce caste composition correlates with local consumer demand composition, and this relationship is significantly *amplified* in food sectors–where caste purity norms are most salient–relative to non-food firms drawing from the same labour pool.

Fact 4: Causal Evidence from Rainfall Shocks
After positive rainfall shocks: (i) LC consumption rises 9.2% relative to HC; (ii) LC-owned firm revenues rise 4.7% relative to HC-owned firms; (iii) HC-owned firms in competitive markets hire 1.6 pp more LC workers; (iv) the effect is stronger in contact-intensive sectors (1.0–1.3 pp). Foreign demand shocks (placebo) produce no differential LC hiring.

Question

"Are these effects economically meaningful?"
Yes. The calibrated model shows these micro-level effects aggregate to a 7.6% reduction in real income from identity barriers. The 21.5% average ad valorem barrier is comparable to the geographic frictions separating Indian states estimated by Asturias et al. (2019). Supply-side hiring frictions alone account for 4.7% of aggregate income–roughly twice the effect of demand-side taste bias.

4 Theoretical Framework

4.1 Model Overview

We develop a Melitz-style heterogeneous-firm trade model in which three caste groups function as segmented markets. Cross-group barriers enter through three channels:

1. **Demand-side taste bias** ($\Psi_{s,s'} = e^{-\beta}$): consumers discount cross-caste varieties
2. **Hiring-cost wedge** ($g_{k,s,s'}$): raises the marginal cost of employing out-group workers
3. **Fixed cost of cross-caste market entry** ($c_{x,s,s'}$): paid in destination labour

4.2 Key Innovation: Destination Labour

A firm from group s selling in market s' produces using group-s' labour. This links demand and supply barriers within the firm's cost structure:

$$\mathcal{C}_{s,s'} = \begin{cases} w_{s'} & \text{if } s = s' \\ (1 + g_{k,s,s'}) \cdot w_{s'} & \text{if } s \neq s' \end{cases}$$

Revenue depends on income above subsistence (non-homothetic preferences), generating differential market sizes:

$$r(z, s, s') = \Psi_{s',s} \left(\frac{\sigma}{\sigma - 1} \cdot \frac{\mathcal{C}_{s,s'}}{z \cdot P_{D,s'}} \right)^{1-\sigma} (1 - a)(I_{s'} - \bar{h})$$

4.3 Composite Caste Resistance

Both frictions collapse into a single iceberg-equivalent cost:

$$T_{s,s'} = \underbrace{e^{\beta/(\sigma-1)}}_{\text{demand bias (7.4\%)}} \times \underbrace{(1 + g_{k,s,s'})}_{\text{hiring wedge (4.7–17.7\%)}}$$

The composite barrier averages 21.5% ad valorem across bilateral routes, ranging from 12.4% (LC→HC) to 26.4% (MC→LC).

Question

"How do you separately identify demand-side versus supply-side barriers?"
Each barrier maps to a distinct set of empirical moments. The taste bias β is identified from the *differential* revenue response of LC versus HC firms to caste-specific demand shocks (Fact 4): if demand were caste-blind, a shift in LC spending would raise all firms' revenues equally. The hiring wedges g_k are matched to worker-composition shares (Fact 1). The fixed costs c_x are matched to hiring participation rates. The mapping is transparent and one-to-one.

5 Quantitative Results & Policy Implications

5.1 Calibration

Calibrated Parameters:

- Taste bias: $\beta = 0.284$, implying consumers discount cross-caste varieties by $\Psi = e^{-\beta} \approx 0.753$
- Hiring wedges: average 13.1% across six bilateral pairs (range: 4.7% to 17.7%)
- Firm-size distribution: Pareto shape $\eta = 3.1$, matched to top-10% revenue share (87.7%)
- 15 parameters total, each matched to a distinct empirical moment

5.2 Key Counterfactuals

Supply-side frictions dominate demand-side bias:

- Removing hiring wedges ($g_k = 0$): +4.7% aggregate income (LC gains most: +6.7%)
- Removing taste bias ($\beta = 0$): +2.6% aggregate income
- Removing both: +7.6% aggregate income, −46% inequality
- The combined gain exceeds the sum of individual effects $\Rightarrow$ the two channels are complementary

Market access is asymmetrically valuable:

- Full autarky: −9.7% aggregate income (LC bears largest loss: −11.4%)
- LC fully isolated: −5.4% aggregate income, nearly all borne by LC
- One-directional barriers (LC can buy but not sell to HC): aggregate income unchanged, but inequality rises 20%–the most regressive form of segmentation

Question

"Is the symmetric taste bias assumption defensible?"
The baseline calibration imposes a common β across all caste pairs, identified from a single differential revenue elasticity. However, the directional counterfactuals in Panel B serve as extreme asymmetric taste bias experiments. Blocking LC→HC trade is equivalent to $\beta_{\mathrm{HC,LC}} \to \infty$. That this extreme asymmetry is primarily regressive (20% rise in inequality) rather than costly in aggregate shows the distributional results are robust to the hierarchy. Moreover, the *total* bilateral barrier–combining taste bias and hiring wedges–is pinned by data moments; removing all identity frictions yields a 7.6% aggregate gain regardless of how the barrier is decomposed between demand and supply sides.

5.3 Robustness of Quantitative Results

- Alternative $(a, \bar{h})$ combinations: aggregate gains range from 5.0% to 9.6%
- Adding spatial iceberg costs $\tau = 1.03$: gains attenuate to 6.1%; at $\tau = 1.10$: to 3.9%
- Attenuation from spatial costs reflects a bounding exercise: if spatial segregation is itself driven by identity barriers, holding τ fixed understates the true gains

Question

"Don't within-group preferences have benefits?"

We acknowledge this limitation explicitly. Identity-based networks may facilitate trust, risk-sharing, intergenerational skill transmission, and information exchange (Cassan et al., 2024; Fisman et al., 2017; Munshi, 2019). Our results measure the *efficiency costs* of identity-based segmentation. The net welfare effect depends on weighing these costs against potential benefits–precisely why we note that "the cost of policies aimed at reducing identity barriers should not exceed the estimated gains."

6 Contribution to Literature

6.1 Primary Contributions

- **Misallocation through identity:** Identifies a novel source of misallocation–identity-based demand and labour-market barriers–and embeds it in a quantitative GE framework that decomposes aggregate losses into demand-side and supply-side channels (complements Hsieh & Klenow, 2009; Boehm & Oberfield, 2020; Hsieh, Hurst & Jones, 2019)
- **Demand-led diversification:** First to document that firms hire cross-caste workers *because of* identity barriers (to access otherwise unreachable consumers), not *despite* them–extends Holzer & Ihlanfeldt (1998), Cook et al. (2023, 2025)
- **Identity and trade:** Separates demand-side taste bias from supply-side hiring frictions and shows the latter dominates quantitatively–advances Fujiy et al. (2025), Boken et al. (2025) on cultural proximity and trade
- **Causal evidence on product-market discrimination:** Uses rainfall shocks to identify taste-based discrimination in product markets, with a foreign-demand placebo ruling out supply-side confounds–builds on Doleac & Stein (2013), Edelman & Luca (2014)

Question

"How does this relate to the emerging literature on customer discrimination and firm hiring?"
Recent work (Cook et al., 2023, 2025; Rubinstein, 2025; Kelley et al., 2024) shows that firms' hiring decisions reflect their customers' biases. We contribute a general-equilibrium model that jointly embeds both demand-side consumer bias and supply-side hiring frictions in the firm's cost structure, allowing us to decompose the macroeconomic and distributional consequences. The key insight is that competition can exacerbate rather than eliminate discrimination when demand segmentation creates incentives for firms to cater to consumer biases through workforce composition.

7 Conclusion

This paper provides causal evidence and a quantitative assessment of how social identity shapes firm growth and workforce composition through consumer preferences. By documenting demand-led workforce diversification and quantifying its aggregate costs, we show that supply-side hiring frictions are the dominant source of misallocation (4.7% of aggregate income), roughly twice the cost of demand-side taste bias. At 21.5% ad valorem, identity-based barriers within local markets are comparable to the geographic frictions separating Indian states–social distance can be as costly as physical distance. Labour-market interventions are likely to yield larger and more equitable gains than policies targeting consumer preferences alone.

For detailed methodology, robustness checks, model derivations, and technical appendices, please see the full manuscript.